

Project Title
Mini Search Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
K. Gayathri Srivalli (2520080010)
M. Hruthika (2520090002)

Team Number: 18
Section: 12

Under the guidance of

Ms. Chandusha Kanda
Asst. Prof, CS&IT



Hyderabad-500090, Telangana, India.

ABSTRACT

This project presents a Mini Search Engine designed as an Information-Retrieval System for efficiently searching and retrieving relevant information from a document corpus. The system implements four query classes: exact keyword search, phrase or substring search, fuzzy search with “did you mean” suggestions, and document-similarity search. A indexing mechanism is developed to organize document terms and enable fast retrieval. Ranking algorithms are applied to prioritize results according to query relevance, term frequency, similarity, and matching quality. Fuzzy matching helps users identify spelling errors and retrieve intended terms, while similarity search discovers documents with related content. All core algorithms, including indexing, searching, ranking, string matching, and similarity computation, are implemented manually to demonstrate algorithmic principles. The project also analyses the time and space complexity of each algorithm and considers scalable design choices for handling larger document collections efficiently. Overall, the system demonstrates applications of data structures, algorithms, and information retrieval concepts.

Signature of the Guide